

Hyperlactemia

Lactic acid is normally produced in excess by about 20 mmol/kg/day, which enters the bloodstream. It is then metabolized mostly via the liver and the kidney. In aerobic conditions, pyruvate is produced via glycolysis and then enters the Krebs cycle, largely bypassing the production of lactate. Under anaerobic conditions, lactate is an end product of glycolysis and feeds into the Cori cycle as a substrate for gluconeogenesis. Elevated lactate is not clearly and universally defined but most studies use cut-offs between 2.0 and 2.5 mmol/L whereas “high” lactate has been defined as a lactate level > 4mmol/L in a number of studies. Furthermore, the “normal value” may vary depending on the assay used.

Lactic acidosis is often used clinically to describe elevated lactate but should be reserved for cases where there is a corresponding acidosis (pH < 7.35).

The exact pathophysiology of elevated lactate in various conditions is likely multifactorial, patient-specific, and disease-specific. In general, lactate elevation may be caused by increased production, decreased clearance, or a combination of both.

The causes of hyperlactatemia is listed in the table 1.1

The following guideline is recommended after reviewing all cases submitted to the metabolic service throughout the period of one year with Haigh lactate:

- 1- Metabolic workup should be initiated by the primary team which includes the following:**
 - TMS(Acylcarnitine+Aminoacid profile)
 - Repeat Lactate, blood gas, ammonia, blood glucoses, CK, LFT and urine dipstick
- 2- Evaluate for other possible cases of hyperlactatemia like hypoxemia, hypovolemia, sepsis, cardiac disease and liver diseases**

When to refer to Metabolic team:

- If the patient has any other abnormal biochemical markers including hypoglycemia, hyperammonemia, metabolic acidosis and hypercekemia
- If there is positive family history of metabolic disorders or sudden death
- If the repeated lactate remained high above 3.5
- If patient noticed to have persistent elevated lactate on multiple occasions
- If the TMS showed abnormal findings

=====

Causes of Hyperlactemia:

Shock

Distributive
Cardiogenic
Hypovolemic
Obstructive

Post-cardiac arrest

Regional tissue ischemia

Mesenteric ischemia
Limb ischemia
Burns
Trauma
Compartment syndrome
Necrotizing soft tissue infections

Diabetic ketoacidosis

Drugs/toxins

Alcohols
Cocaine
Carbon monoxide
Cyanide

Pharmacological agents*

Linezolid
Nucleoside reverse transcriptase inhibitors
Metformin
Epinephrine
Propofol
Acetaminophen
Beta₂ agonists
Theophylline

Anaerobic muscle activity

Seizures
Heavy exercise
Excessive work of breathing

Thiamine deficiency

Malignancy

Liver failure

Mitochondrial disease

Table 1.1

References:

- 1- **Lactic Acidosis**, Chris D. Foucher; Robert E. Tubben.
- 2- **Etiology and therapeutic approach to elevated lactate**, Lars W. Andersen, BS, Julie Mackenhauer, MD.